

DEB theory's tetrahedron of life links feeding and reproduction to trophic position of species^{☆,☆☆}

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ARTICLE INFO

Keywords:

Dynamic energy budget theory
Over-exploitation avoidance
Supply–demand spectrum
Altricial-precocial spectrum
Life-time cumulated neonate mass production

ABSTRACT

Dynamic Energy Budget (DEB) theory's tetrahedron of life links the inner to the outer world of individuals, beyond energy and mass conservation. It follows mathematically from the some assumptions behind DEB theory, without using new assumptions and links the supply stress, the fraction of assimilation to reproduction and the minimum scaled functional response to reach puberty all to the fraction of mobilized reserve to soma: the cornerstone of life. The supply stress quantifies the supply–demand spectrum: supply-species eat what is available, while demand-species eat what they need. The minimum scaled functional response to reach puberty increases with the supply stress, which implies that the half-saturation coefficient needs to decrease with the supply stress to ensure that the feeding rate is close to its maximum for species near the demand end of the supply–demand spectrum. This coefficient is the ratio of the specific ingestion and food searching rates, so the food searching rate needs to increase with the supply stress. This is in harmony with the recent finding that the factorial aerobic scope increases from 3.2 for zero supply stress to 32 for maximum supply stress; a high food searching rate requires a high peak metabolism. It turns out that the supply stress needs to be low for the allocation to reproduction to be large. The supply stress has direct links with the trophic position and the precociality index, i.e. the ratio of the maturity levels at birth and puberty. The lower the fraction of assimilation to reproduction, the lower the life-time cumulated neonate mass production as well as the precociality index within a variety of taxa within the Add_my_Pet (AmP) collection of energetic data and DEB parameters. The intra-class variation of the precociality index with the fraction of assimilation to reproduction turns out to differ from the inter-class variation. Invertebrates were found to have a low supply stress, have a low trophic position, allocate a large fraction of their assimilation to reproduction and make relatively small eggs. Tetrapods, especially birds and mammals, have a high supply stress, a high trophic position, invest little in reproduction and their neonates are relatively large. This coupling of traits follows from the structure of DEB theory and is empirically confirmed by the 7334 animal species in the AmP collection for a wide range of taxa. Species with a high trophic position must have a food intake close to their maximum and cannot invest much in reproduction. The combined implications for feeding and reproduction avoids or reduces over-exploitation of resources in the food web and might contribute to the dynamic stability of food webs and to the maintenance of biodiversity in ecosystem models, challenging fitness maximization as central theme in evolution biology. The tetrahedron provides strong support for the central position of the supply–demand spectrum in animal energetics.

[☆] This article is part of a Special issue entitled: 'DEB2025' published in Ecological Modelling.

Article info

Supporting info: <https://github.com/add-my-pet/SI/tree/main/Kooy2026a/>.

Supporting code: https://github.com/add-my-pet/DEBtool_M and AmPtool.

Database link: https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/.

The SI has the code to generate the figures in this paper, using the AmP data base.

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<https://doi.org/10.1016/j.ecolmodel.2026.111771>

Received 25 February 2026; Received in revised form 9 July 2026; Accepted 27 July 2026

Available online 4 August 2026

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1. Introduction

Food web dynamics is key to systems ecology centred around the possibly positive relationship between complexity and stability (Landi et al., 2018) and inverse relationship between species diversity and productivity (Huston, 1994); foraging theory is an important aspect of food web dynamics (Moosmann et al., 2021). Population densities of species, and their reproduction rates, tend to decrease for increasing trophic position; fluctuations in population density at low trophic levels can lead to local extinctions at high trophic levels (Huston, 1994). The trophic position of species in food webs is traditionally linked to body size, based on the idea that large-bodied species feed on smaller-bodied ones (Kopf et al., 2021). Yet, these links are complex (Arim et al., 2007; Deleambre et al., 2026; Jia et al., 2024; Nakazawa et al., 2010) and common phenomena such as intraguild predation makes the concept ‘trophic position’ less than exact (Arim and Marquet, 2004). More recently, the trophic position is derived from isotope dynamics, which involves strong assumptions about metabolism (Doi et al., 2012; Riekenberg et al., 2021; Emmery, 2012; Chikaraishi et al., 2014; Post, 2002; Kjeldgaard et al., 2021). These assumptions concern particular aspects of metabolism that relate to isotope distributions, not to the general energy metabolism. This paper focuses on the links between the trophic position and the general energy metabolism as implied by Dynamic energy Budget (DEB) theory, with an important roles for foraging behaviour and other life history aspects.

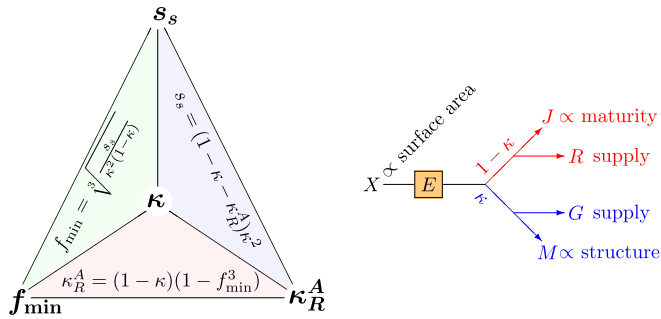


Fig. 1. Left: The tetrahedron of life. Right: Allocation in the std DEB model.

κ	fraction of mobilized reserve to soma
X	food
M	somatic maintenance
κ_R^A	fraction of assimilation to reproduction
E	reserve
J	maturity maintenance
$f_{\min}$	min scaled func resp to reach puberty
G	growth
R	maturation/reproduction
s_s	supply stress $\dot{p}_J \dot{p}_M^2 / \dot{p}_A^3$, with $\dot{p}_J$ maturity maint, $\dot{p}_M$ somatic maint, $\dot{p}_A$ assimilation

The main task of energy budget models for animals is to quantify how much food is eaten, and how many offspring are produced during ontogeny until death by aging. Yet this task is more complex than it might seem at first sight. One complicating factor is the budget aspect, which means that the various metabolic fluxes within an individual are very much interlinked: any flux depends on all the rest. So the processes of feeding, digestion, defecation, urination, drinking, respiration, heat dissipation, assimilation, storing, allocation, growth, reproduction, development, somatic and maturity maintenance and aging all have to be studied simultaneously and in coherence in dynamic environments, while the resulting model should be simple enough to apply in practice.

Another recently emerging factor that complicates energy budget models is that parameter values not only have a strong taxonomic coherence (Lika et al., 2023), but also depend on the trophic position of species in the food web (Kooijman, 2026; Verhille and Kooijman, 2026). Moreover, one particular parameter, the specific somatic

maintenance, varies from 0.5 to 30831 J/d cm³ at 20°C among the Add_my_Pet (AmP) species; its actual value turns out to depend on the seasonal dynamics of food (Lika et al., 2026). Growth is intimately linked to somatic maintenance. The von Bertalanffy growth is implied by DEB theory if food density is constant, and has a detailed metabolic interpretation in the DEB context. It is proportional to the specific somatic maintenance and 1.5 times the von Bertalanffy growth rate has the interpretation of the specific growth rate at maximum growth (Lika and Kooijman, 2024). So the context matters, which has implications for all aspects of energy budget models, including the data that is required for estimation of parameters and the evaluation of the resulting parameter estimates.

DEB theory aims to specify a set of related energy budget models for all organisms in a way that explicitly respects energy and mass conservation on the basis of first principles, not using optimization arguments or allometric functions. The standard (std) DEB model A_G^R for animals in Fig. 1 is the only one among the 21 topological alternatives that is consistent with a set of stylized facts (Lika and Kooijman, 2011). The explicit implementation of conservation laws strongly restricts alternatives (Kooijman, 2020b) and makes homeostasis assumptions unavoidable. The AmP collection (Marques et al., 2018) demonstrates that DEB models are simple enough to apply in practice, even for extinct species (AmP, 2026). This rich database invites a search for patterns in parameter values to further develop meta theory for DEB theory. The search for patterns was successful indeed and recently resulted in the insight that supply–demand spectrum (Lika et al., 2014; Verhille and Kooijman, 2026) is key to the interaction of the energy budget with the environment, not only in terms of behaviour and food acquisition, but also in trophic relationships between species.

The aim of this paper to analyse the interaction between the inner and outer worlds of individuals, separated by the gut wall, counting gut contents as outer world. Metabolism is dominant in the inner world, while in the outer world, the trophic position of species plays an important role, so the ecological setting of both prey/food and predator.

The structure of the paper follows the ideas on the methodology of theories and meta theories of Kooijman et al. (2024): Theories are derived from a set of assumptions, which should be tested each empirically; this has been done for DEB theory, which resulted in a large number of papers. Meta theories are, in this case, are about ideas behind the co-variation of parameter values, or traits that are functions of these parameter values. These meta theories again require assumptions, but these assumptions do not directly relate to assumptions on which the theory itself is based. This second set of assumptions behind meta theory might be falsified, without impacting the first set of assumptions behind the theory itself. The next section specifies DEB theory’s tetrahedron of life and the reason for its name. Since it directly follows mathematically from the assumptions behind DEB theory, it is part of DEB theory itself, not of meta theory. The results section shows the relationship between the fraction κ_R^A of assimilation that is allocated to reproduction and the specific life-time cumulated neonate mass production, and the precociality index. These results are part of meta theory, which is why it follows after a short methods section. The general discussion section provides further context.

2. Tetrahedron of life

Supply-species eat what is available and are very flexible in the ontogeny of growth and reproduction, while demand-species eat what they need and are much less flexible in growth and reproduction, following rather fixed patterns, but are more advanced in food acquisition (Lika et al., 2014; Verhille and Kooijman, 2026). The supply stress $s_s = \dot{p}_J \dot{p}_M^2 / \dot{p}_A^3$ quantifies the supply–demand spectrum, where $\dot{p}_A$, $\dot{p}_J$ and $\dot{p}_M$ stand for the energy fluxes assimilation, maturity and somatic maintenance. Although this quantity can be evaluated at any point in the ontogeny of the individual after birth, it is here applied

to fully-grown individuals at constant food and temperature. Many metabolic and eco-physiological properties of species are associated with the supply stress, such as the modes of embryo development, the occurrence of capillaries in demand species and the capacity to handle periods of starvation. The fraction $1-\kappa$ of mobilized reserve is allocated to maturity maintenance plus maturation (in juveniles) or reproduction (in adults), see Fig. 1. Allocation to maturity and somatic maintenance is demand-organized, meaning that it does not depend on substrate availability, while allocation to maturation (or reproduction) and to growth are supply-organized, meaning that it does depend on substrate availability, i.e. the left-over after allocation to maintenance. Extreme starvation can cause rejuvenation (reduction of maturity) and shrinking of structure, so modifies this allocation

The concepts maturity and structure are unique for DEB theory, as far as I know (Kooijman, 2020b); see Fig. 1. The supply stress is independent of size, both ontogenetically and inter-specifically. This is because of some key DEB assumptions: assimilation is proportional to surface area, maturity maintenance is proportional to maturity and somatic maintenance is proportional to structure. The relations for assimilation and somatic maintenance go at least back to A. R. Wallace (1865), see Finch (1990, App 3) and they are used by Pütter (1920) to model growth, see Kearney (2021). The supply stress popped up in the search for the physical boundaries of the parameter space in the std DEB model (Lika et al., 2014); constraints that are now used routinely to stabilize parameter estimation in the AmP collection. Beside general conditions, such as parameters cannot become negative, the most stringent condition turns out to be that it is necessary that puberty can be reached. This is only the case if the scaled function response is larger than some minimum value $f_{\min}$, and $f_{\min} < 1$ must apply.

Fig. 2 shows the surface of $(s_s, \kappa, f_{\min})$, with the minimum scaled functional response to reach puberty $f_{\min} = \sqrt[3]{\frac{s_s}{\kappa^2(1-\kappa)}}$, see Lika et al. (2014); it also shows the surface of $(s_s, \kappa, \kappa_R^A)$, with the fraction of assimilation to reproduction $\kappa_R^A = 1 - \kappa - s_s/\kappa^2$, see Verhille and Kooijman (2026). These equations result from the structure of DEB theory, no empirical argument is involved, no extra assumptions added to the list of DEB assumptions, no new parameters are introduced.

The derivation of these two equations only uses (1) the allocation scheme of Fig. 1, (2) the strong homeostasis assumption (the chemical composition of reserve and structure remains constant), (3) isomorphy (surface area $\propto$ squared length and structure $\propto$ cubed length), (4) further allocation to maturation ceases and that to reproduction starts at puberty and (5) constant food density. The latter requirement is only used for the derivation. The expression for κ_R^A still holds for slowly varying food densities; assimilation responds instantaneously to changes in food density, but allocation to reproduction is smoothed out by reserve (and the reproduction buffer).

If maturation and growth cease at puberty for a individual at constant food, we have allocation to reproduction is zero, $\dot{p}_R^\infty = 0$, while $\dot{p}_R^\infty = (1-\kappa)\dot{p}_C^\infty - \dot{p}_J^\infty$, where $\dot{p}_C^\infty$ is the reserve mobilization rate and $\dot{p}_J^\infty$ is the allocation to maturity maintenance (all powers $\dot{p}_*$ have units J/d). This equals zero if $(1-\kappa)\dot{p}_C^\infty = \dot{p}_J^\infty$. Assimilation equals $\dot{p}_A = f\dot{p}_{Am}$, where $\dot{p}_{Am}$ is the maximum assimilation rate and f the scaled functional response. Absence of growth implies that reserve does not change at steady state $\dot{p}_C^\infty = \dot{p}_A = f_{\min}\dot{p}_{Am} = \dot{p}_M^\infty/\kappa$. Given that s_s is defined as $s_s = \frac{\dot{p}_J^\infty \dot{p}_M^\infty}{f_{\min}^3 \dot{p}_{Am}^3}$, substitution of $\dot{p}_J^\infty$, $\dot{p}_M^\infty$ and $\dot{p}_A$, gives $s_s = f_{\min}^3 (1-\kappa)\kappa^2$, so $f_{\min} = \sqrt[3]{\frac{s_s}{\kappa^2(1-\kappa)}}$.

For a fully grown individual at constant food (allocation to growth being zero), the allocation scheme of Fig. 1 implies $\kappa_M^A + \kappa_J^A + \kappa_R^A = 1$, where $\kappa_M^A = \frac{\dot{p}_M}{\dot{p}_A}$ is the somatic maintenance as fraction of assimilation, $\kappa_J^A = \frac{\dot{p}_J}{\dot{p}_A}$ is the maturity maintenance as fraction of assimilation and $\kappa_R^A = \frac{\dot{p}_R}{\dot{p}_A}$ is the reproduction power as fraction of assimilation. Since $\kappa_M^A = \kappa$ (else growth or shrinking occurs) and supply stress s_s is defined as $s_s = \frac{\dot{p}_J \dot{p}_M^2}{\dot{p}_A^3} = \kappa_J^A \kappa_M^A$, we directly have $\kappa_R^A = 1 - \kappa - s_s/\kappa^2$.

The third equation of the tetrahedron, $\kappa_R^A = (1-\kappa)(1-f_{\min}^3)$, directly follows from the two other ones and points to the direct connection between reproduction investment and feeding. The hidden underside of the tetrahedron, i.e. the side that is independent of κ and only depends on s_s , $f_{\min}$ and κ_R^A , has equation $s_s(1-f_{\min}^3)^3 = \kappa_R^A(1-f_{\min}^3 - \kappa_R^A)^2$; it follows also from the two equations.

The tetrahedron is called the tetrahedron of life because it brings many aspects of energy budgets into play: basically assimilation, reproduction, maturity and somatic maintenance, but via these quantities many more aspects are involved. More specifically the tetrahedron links feeding and reproduction, so key aspects of the interaction between the individual with its environment, to the trophic position of species, so its ecological setting; it shows the central role of κ , which can be called the cornerstone of life. An ecologically important property of the tetrahedron is that the supply stress needs to be small for the fraction of assimilation that is allocated to reproduction to be large. This explains why species at the bottom of the food chain need to have a small supply stress to compensate the predatory losses via reproduction.

Fig. 2 illustrates the $(s_s, \kappa, f_{\min})$ - and the $(s_s, \kappa, \kappa_R^A)$ -surfaces. The isocline for $f_{\min} = 1$ or $\kappa_R^A = 0$, the fat black curve $s_s(\kappa) = \kappa^2(1-\kappa)$, follows from the condition that it must be possible to reach puberty, or else reproduction cannot take place.

The left panel shows that the minimum scaled functional response to reach puberty increases with the supply stress. DEB theory uses Synthesizing Units dynamics (Kooijman, 1998) to specify how the feeding rate depends on food density. In its most simple form this dynamics reduces to the well-known Holling type II model (Holling, 1959), which has the half-saturation coefficient as parameter. Since food density generally varies in time, the implication is that the half-saturation coefficient needs to be small for species with a high supply stress, to ensure that food intake is close to its maximum, even for (rather) low food densities. DEB's half-saturation coefficient is the ratio of the surface area specific maximum ingestion rate and surface area specific food searching rate; it can only be small if the specific food searching rate is large. This is in beautiful harmony with the finding that the factorial aerobic scope (FAS) increases with the supply stress. Verhille and Kooijman (2026) propose the one-parameter empirical model $10 \log(\text{FAS}/\text{FAS}_s) = s_s/s_s^d$, with $10 \log \text{FAS}_s = 1/2$, being the log of FAS at the supply-end of the spectrum, and $s_s^d = 4/27$ is the maximum value of s_s at the demand-end of the spectrum; so FAS increases from 3.2 for $s_s = 0$ to 32 for $s_s = 4/27$. Active food searching typically involve spells of high activity and with a high FAS, the individual is able to do what is needed. The expectation that the half-saturation coefficient decreases for increasing supply stress is hard to test directly against data, in view of the diversity of habitats, prey species and food searching methods. While extreme supply species can handle to full range of scaled functional responses, (0,1), to reach puberty, extreme demand species need maximum food intake. Notice that the isoclines for $f_{\min} = 0$ and $f_{\min} = 1$ meet in points $(s_s, \kappa) = (0, 0)$ and $(s_s, \kappa) = (0, 1)$. It is no coincidence that birds and mammals, i.e. the taxa with the highest supply stress, typically exhibit metabolic up-regulation of assimilation during reproduction events.

The right panel of Fig. 2 shows that the isocline for $\kappa_R^A = 1$ is reduced to the point $(s_s, \kappa) = (0, 0)$, but no species can be in that neighbourhood. The edge for low κ , lower than the $\kappa(s_s)$ -curve for which κ_R^A is at maximum, is poorly populated by species, especially for low s_s . The species that presently come closest in the AmP collection is the salp *Thalia democratica* with $(s_s, \kappa, \kappa_R^A) = (2.7 \cdot 10^{-7}, 0.1, 0.9)$ and the Alaska plaice, *Pleuronectes quadrituberculatus*, with $(s_s, \kappa, \kappa_R^A) = (6.4 \cdot 10^{-5}, 0.0687, 0.91)$. The maximum value of κ_R^A for the AmP species is 0.9, but few species have such high values. The 95th percentile is 0.46, and the median value is 0.045; see the general discussion section.

Verhille and Kooijman (2026) and Lika et al. (2019) showed that κ and κ_R^A follow beta distributions accurately among the species, and s_s a truncated beta distribution, i.e. truncated to the interval (0, 4/27); this explains the uneven distribution of the AmP-points in Fig. 2.

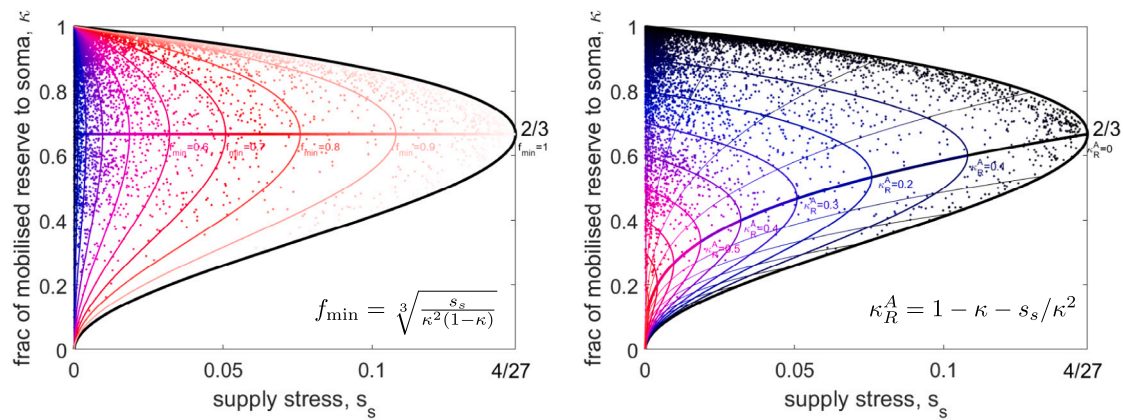


Fig. 2. The isoclines $\kappa(s_s)$ of the minimum values of f to reach puberty (left) and of the fraction of assimilation that is allocated to reproduction (right) and the values of $\kappa(s_s)$ for which $f_{\min}$ and κ_R^A are at maximum, i.e. $\kappa(s_s) = 2/3$ and $\sqrt[3]{2s_s}$, respectively. The values of the points $(s_s, \kappa, f_{\min})$ and $(s_s, \kappa, \kappa_R^A)$ for the species in the AmP collection at 2026/02/26 are indicated. All mentioned quantities are dimensionless.

While the left panel is key to the link between supply stress and FAS, the right panel is key to the link between supply stress and the trophic position of species. The first hints came from [Lika et al. \(2014\)](#): invertebrates and ray-finned fish have a low supply stress, cartilaginous fish, amphibians and reptiles score moderately and birds and mammals have a high supply stress. These findings are further supported by [Verhille and Kooijman \(2026\)](#) and [Kooijman \(2026\)](#): rodents were found to have lower supply stress than carnivores and typically have a lower trophic position and are more fecund. Likewise, the supply stress was found to increase in the aquatic environment from crustacea, to clupeiformes (herrings) to selachii (sharks) and in the terrestrial environment from hexapoda (insects), to invertebrate-eating passeriformes (song birds) to hieraves (birds of prey), which typically occupy trophic positions 1, 2 and 3 in food chains ([Kooijman, 2026](#)). Ray-finned fish have many tiny neonates and play an important role in marine food webs; their low supply stress, so high κ_R^A , is no coincidence.

3. Methods

All data in the next section come from the AmP collection and represent (functions of) DEB parameters for animal species, not directly measured values. Apart from taxonomic coherence, five different patterns in the co-variation of DEB parameter values are identified so far ([Augustine et al., 2019a,b](#); [Kooijman, 1986, 2013, 2014, 2026](#); [Lika et al., 2019, 2023, 2026](#); [Marques et al., 2018](#); [Verhille and Kooijman, 2026](#)). Although these patterns greatly contribute to biological understanding and are key to meta theory for DEB theory ([Kooijman et al., 2024](#)), they also show that DEB parameters and functions of these parameters are statistically dependent. Statistical tests typically assume statistical independence of data. I think that presenting statistical tests based on assumptions that are known to be violated is really misleading, which is why no such tests are done in this paper on the co-variation of parameter values.

Given that life history aspects play important roles in this paper, a potential source of disappointment of readers might be that little attention is given to current life-history theory. The reasons are (1) the coupling of traits of individuals in this paper is specific for DEB theory, (2) any coupling of traits requires a model for energy budgets, but I am unaware of such a model in the literature on life history theory, (3) life history papers typically apply optimization arguments and use constant conditions and homogeneous space, which do not exist in practice ([Kooijman, 2004](#)). DEB theory avoids the use of optimization arguments, see the general discussion.

Since the tetrahedron is a mathematical implication of DEB models, the empirical support for it is basically the empirical support for DEB theory and models. The median relative error of predictions for

underlying data in the AmP collection ([Marques et al., 2018](#); [AmP, 2026](#)) is 0.039, for in total 7334 animal species, 74745 data sets and 645299 data points at 2026/04/20; the entries were written by 215 authors. Apart from the taxonomic coherence, five patterns in parameter values ([Lika et al., 2023](#)) and over 1330 successful applications and tests for realism of DEB theory are further empirical support for DEB models ([DEBlibrary, 2026](#)); the publications were written by some 2500 authors. DEB models are frequently used in ecotoxicity research ([Kooijman, 2018](#)), where toxicants affect particular parameters; these perturbations are typically well-captured. Measurements on e.g. growth and reproduction of test animals are used to identify which parameters changed: the mode of action of the tested chemical compound. Since the indirect calorimetry model of [Lavoisier and Laplace \(1780\)](#), some 30 popular empirical models for very different aspects of metabolism turned out to be special cases of DEB models, or very good numerical approximations ([Kooijman, 2010b](#), Table 11.1); each of them has extensive support by data. DEB theory shows in what respect these models are connected, provides mechanistic underpinning and specifies conditions for which these models can be expected to hold. The tetrahedron points to links between supply stress with the half-saturation coefficient and with the trophic position. The empirical support for these links was discussed in the previous section.

The next section discusses how the fraction κ_R^A of assimilation to reproduction relates to the integrated reproduction investment and to the precociality coefficient. It shows that κ_R^A is proportional to the specific life-time cumulated neonate mass production ω_R , which involves the aging process; the longer an individual lives, the more neonate mass accumulates. The aging process directly relates to respiration, at least in the DEB context, and maintenance and various overheads contribute to respiration. The section also shows that κ_R^A is also proportional to the precociality index s_H^{bp} : the maturity at birth as fraction of the maturity at puberty, i.e. the maximum value of maturity.

4. Results and discussion

[Fig. 3](#) shows that mass-specific life-time cumulated neonate mass production ω_R is proportional to the fraction κ_R^A of assimilation that is allocated to reproduction: $\int_0^{a_m} \dot{R}(t) W_w^b / W_w^\infty dt = \alpha \kappa_R^A$, where a_m is age at death, $\dot{R}(t)$ is the reproduction rate at time t since birth and α a proportionality constant. The factor W_w^b / W_w^∞ accounts for the large differences in (absolute) neonate size among the taxa; the neonates of molluscs and ray-finned fish are tiny, compared to cartilaginous fish and mammals. [Augustine et al. \(2022\)](#) showed that the life-time cumulated neonate mass production roughly equals the ultimate weight, implying that the weight-specific one would be about equal to one, so the log is about equal to 0. This is consistent with [Fig. 3](#) which shows

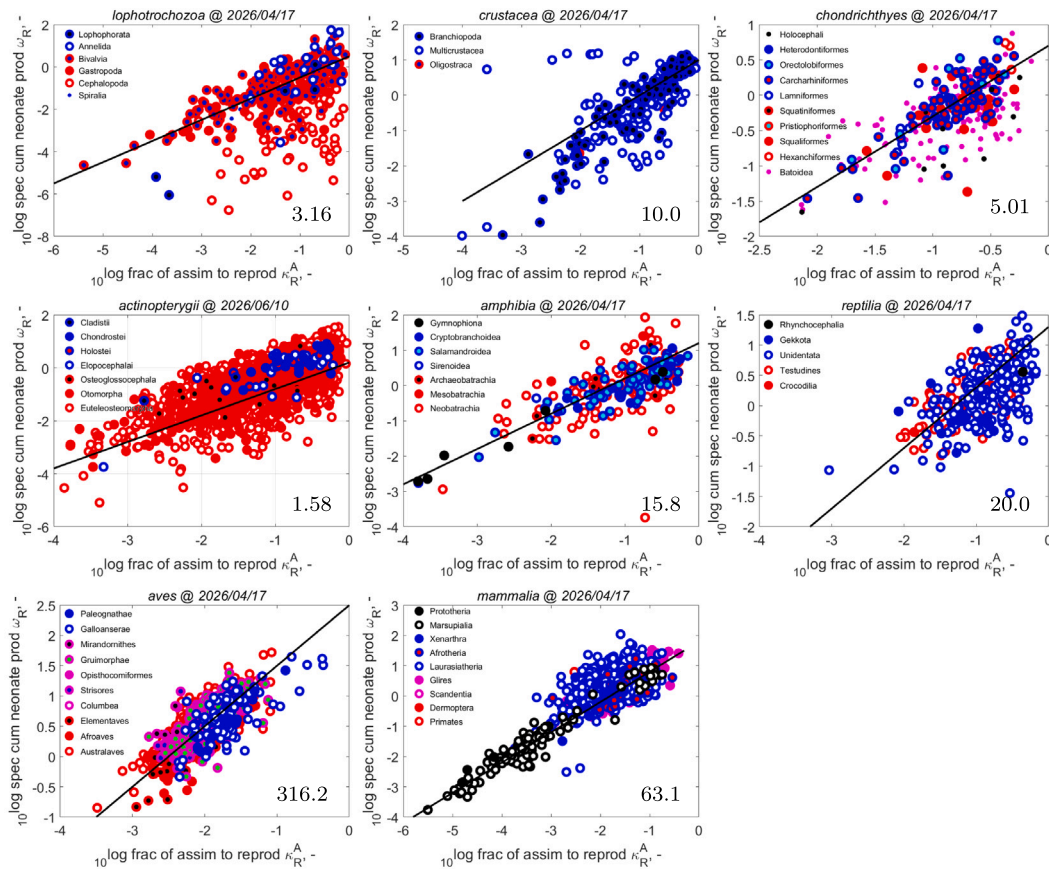


Fig. 3. The specific life-time cumulated neonate mass production, $\omega_R = \int_0^a \dot{R}(t) W_w^b / W_w^\infty dt$, as function of the fraction κ_R^A of assimilation that is allocated to reproduction for various taxa. All lines have slope one and represent $\omega_R = \alpha \kappa_R^A$; the values for α are given in the graphs. Data from the AmP collection. Number of species: 734 Lophotrochozoa, 214 Crustacea, 240 Chondrichthyes, 2790 Actinopterygii, 245 Amphibia, 371 Reptilia, 1741 Aves, 1203 Mammalia.

scatterings around the ordinate value 0. The present finding can be seen as a further refinement that reduces scatter substantially and links production to supply stress.

The proportionality constant α ranges from 1.58 for ray-finned fish (actinopterygii) to 316.2 for birds (aves). The value of α links to the mean supply stress for that taxon, where invertebrates (lophotrochozoans and crustaceans) and ray-finned fish score low, so allocate a lot to reproduction, and endotherms (birds and mammals) score high, so allocated little to reproduction. Consistent with Verhille and Kooijman (2026), birds are found to invest very little in reproduction, relative to other taxa and compensate this by living long; most of their reproduction investment is not in egg production, but in parental care. Parental care is part of somatic maintenance. Treating a very small fraction of somatic maintenance as maturity maintenance to account for parental care avoids that any bird has a low supply stress (Verhille and Kooijman, 2026, App C), see Kooijman (2026). This correction hardly affects somatic maintenance, but strongly affects the supply stress. The cephalopods also invest little in reproduction, but for a different reason: most cephalopod species are semelparous and exhibit programmed death, escaping the consequences of aging (Kooijman and Augustine, 2022).

The relative size at birth is dominated by the precociality coefficient $s_H^{bp} = E_H^b / E_H^p$ (Augustine et al., 2019a), where E_H^b and E_H^p are the maturity levels at birth and puberty, respectively. Birth is defined as the moment at which feeding starts and puberty is the moment at which maturation is completed and allocation to reproduction starts. Since the maturity density, i.e. the maturity per structural volume, does not vary that much, the precociality coefficient s_H^{bp} is about proportional to the weight ratio W_w^b / W_w^p , see Fig. 4. The relevance for food chain dynamics is that, apart from total mass, numbers are important: individuals are

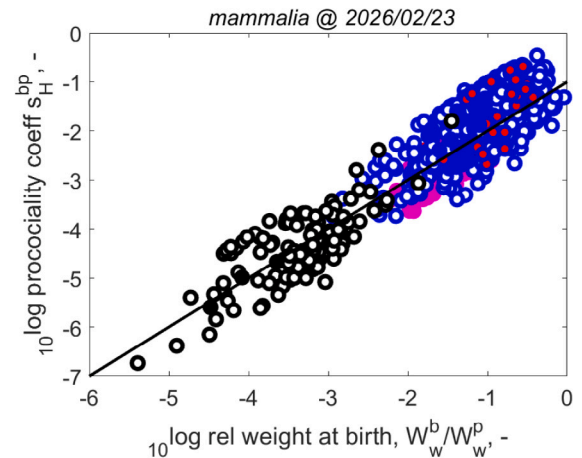


Fig. 4. The precociality coefficient for mammals as function of the ratio of the weights at birth and weights at puberty. The line has slope 1 and represents $s_H^{bp} = 0.1 W_w^b / W_w^p$. Markers as in Fig. 3 for mammals.

the target for selection, not mass. The hazard rate typically drops steeply with increasing age, especially when the neonates are tiny (Kooijman, 2024).

The results for life-time cumulated neonate mass production strongly suggest that species have the evolutionary choice between many tiny neonates or a few larger ones. Tiny neonates typically live

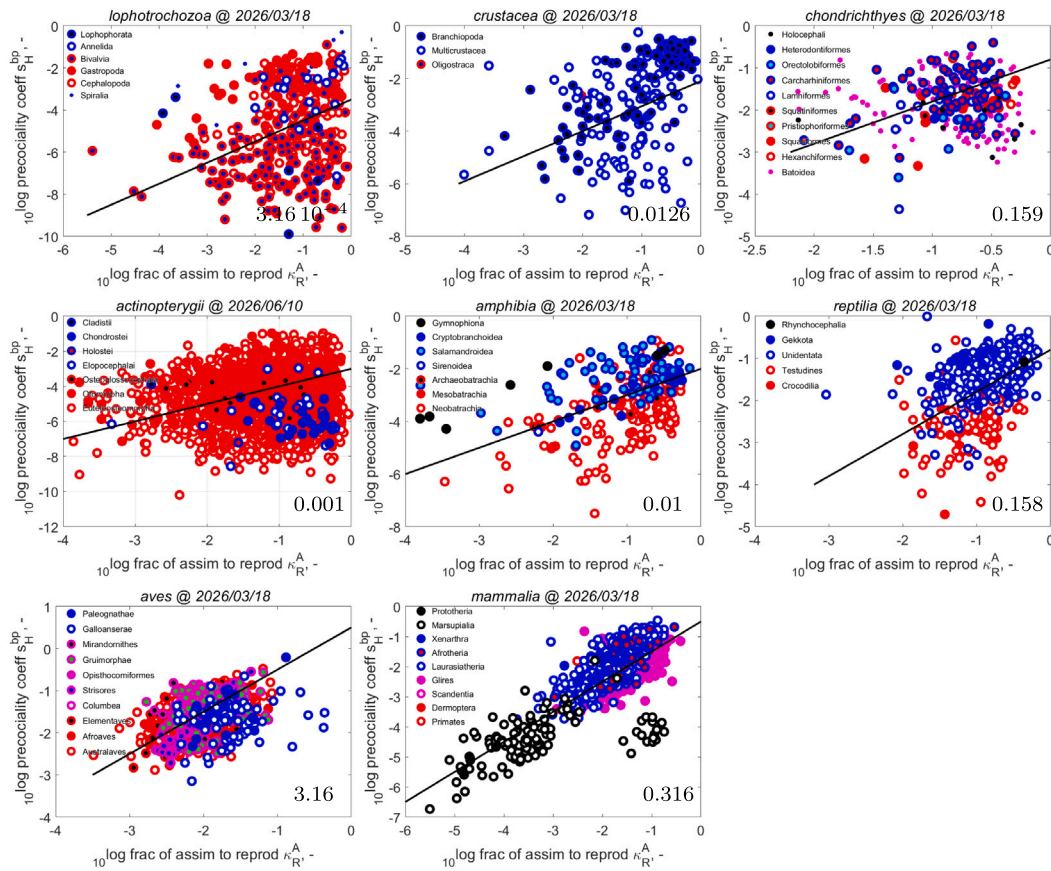


Fig. 5. The precociality index $s_H^{bp} = E_H^b/E_H^p$ as function of the fraction κ_R^A of assimilation that is allocated to reproduction for various taxa. All lines have slope one and represent $s_H^{bp} = \beta \kappa_R^A$; the values for β are given in the graphs. Data from the AmP collection. Number of species: 734 Lophotrochozoa, 214 Crustacea, 240 Chondrichthyes, 2790 Actinopterygii, 245 Amphibia, 371 Reptilia, 1741 Aves, 1203 Mammalia.

in other (micro) habitats than their parents, frequently exhibit metamorphosis and live of different food types. This weakens the concept ‘the trophic position’ for a species, since this changes during ontogeny. The large numbers of tiny neonates directly link to risk management. Petrels (Procellariiformes) typically breed on small ocean islands and produce a single egg that is relatively large for birds (Kooijman, 2020a, 2010b). Although large eggs need long incubation, they reduce the required nesting time, i.e. the period that the parents need to return to that island till the fledge of the young, for a given maturity at fledge. Petrels use very large sea surface areas for foraging; the need to return frequently to one particular island reduces their foraging options. The holo-metabolic insects have $s_H^{bp} = 1$, so birth coincides with puberty and they start allocation to reproduction directly after birth. The larva accumulates this investment and the imago distributes the eggs after pupation (Llandres et al., 2015). Many species do not eat as imago, or only take nectar, which extends the life span of the imago, but does not contribute to egg production. Some species, however, do take protein-rich food that supports egg formation. These examples illustrate that the detailed understanding of the altricial-precocial spectrum (Augustine et al., 2019a) is taxon-specific, which makes the next finding even more remarkable since it is not taxon-specific.

Fig. 5 shows that the precociality coefficient s_H^{bp} is about proportional to the fraction κ_R^A of assimilation that is allocated to reproduction: $s_H^{bp} = \beta \kappa_R^A$. The proportionality constant β ranges from $3.16 \cdot 10^{-4}$ for lophotrochozoa, with molluscs as main taxon, to 3.16 for birds (aves); the values links to the typical relative mass of neonates, compared to the fully grown adults; the smaller β , the smaller the neonate. The high value for birds is possibly an adaptation to reduce the nesting period, so the need for parental care. Mammals most clearly show that the smaller κ_R^A , the higher trophic position, the larger s_H^{bp} . This

seems to be a compensation to reduce risk; with a low s_H^{bp} , they make relatively small offspring, and can still produce a few of them with a low allocation to reproduction. This compensation behaviour might be more important for mammals and birds as demand taxa than for other taxa. The relationship seems almost absent in the lophotrochozoa and the multicrustacea, but in the branchiopods and the ray-finned fish, which also produce relatively small eggs, the relationship still seems to hold. If κ_R^A is high, the species has a low trophic position, so experience a high predation pressure and needs to invest a lot in reproduction to balance.

Fig. 6 shows that the proportionality constants in Figs. 3 and 5 are very much related; the precociality coefficient as fraction of κ_R^A , s_H^{bp}/κ_R^A , is about proportional to the squared mass-specific life-time cumulated neonate mass production as fraction of κ_R^A , so $s_H^{bp}/\kappa_R^A = 10^{-4} (\omega_R/\kappa_R^A)^2$ or $s_H^{bp} = 10^{-4} \omega_R^2/\kappa_R^A$. Figs. 3 and 5 show that ω_R as well as s_H^{bp} are proportional to κ_R^A , so the expectation is that s_H^{bp} is proportional to ω_R , not to ω_R^2 . These figures compare class by class, while Fig. 6 compares all classes/taxa simultaneously. The intra-class variation appears to differ from the inter-class variation. Augustine et al. (2022) found that the scaling exponents of birth weight as function of ultimate weight, differs quite a bit between the vertebrate classes. The Supporting Information shows that this scaling exponent is 0 for ray-finned fish, 0.2 for amphibia, 0.4 for crustaceans, 0.6 for reptiles, 0.8 for birds, close to 1 for mammals, cartilaginous fish and lophotrochozoans. The latter taxon has a large scatter; the sub-taxa are very different. The scaling exponent of puberty weight is close to one for all taxa. Although the answer to the “why” question for the variation of the precociality index is still elusive at the moment, the fact that a relationship clearly exists is truly remarkable.

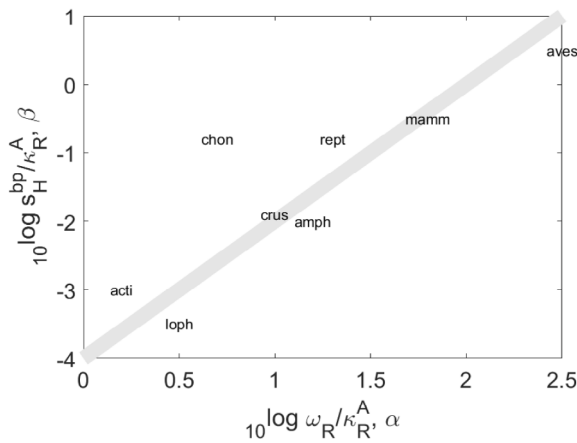


Fig. 6. The precociality coefficient s_H^{bp} as function of the mass-specific life-time cumulated neonate mass production ω_R , both divided by the fraction κ_R^A of assimilation to reproduction, for the taxa in Figs. 3 and 5: lophotrochozoa, crustacea, chondrichthyes, actinopterygii, amphibia, reptilia, aves, mammalia. The data are the proportionality constants α and β in both figures. The line represents the relationship $s_H^{bp}/\kappa_R^A = 10^{-4} (\omega_R/\kappa_R^A)^2$ or $\beta = 10^{-4} \alpha^2$.

5. General discussion

DEB's tetrahedron of life essentially links the trophic position of a species in the food web to the metabolic machinery behind the energy budget, with implications for food searching. The supply–demand spectrum plays a key role and the quantifier supply stress is a simple function of assimilation, maturity and somatic maintenance. To my knowledge, no other theory or model uses the concept maturity maintenance; it includes maintenance of metabolic functions relating reproduction, but also of defense systems, such as the immune system, to capture that poor nutritional conditions make the individual more vulnerable for pathogenic attack. It is tempting to link somatic maintenance to basal metabolic rate as respiration flux, avoiding contributions from various overheads in resting situations of the subject. Yet this strategy is prone to errors for two reasons. First it might be hard to fully avoid overheads, such as growth overheads, without affecting the health of the individual and maturity maintenance cannot be avoided and is not an overhead. The second reason is how the size of the individual is taken into account. The standard way to do this is using weight, but weight has contributions from reserve, which does not require maintenance, and possibly also from the reproduction buffer. The contribution of reserve to weight is widely accepted through the use of the condition factor, weight over cubed length (Heincke, 1908). While specific respiration clearly decreases with weight, the volume-specific somatic maintenance depends much less clearly on weight, as shown in Lika et al. (2026). Therefore the partitioning of biomass into reserve and structure is basic to the comparison.

Charnov et al. (2001) reported that the quantity $(a_m - a_p) \dot{R}_\infty W_w^b / W_w^\infty$ is approximately invariant across species, where a_p is age at puberty and $\dot{R}_\infty$ the ultimate reproduction rate. The median value of the ratio of length at puberty and ultimate length, L_p/L_∞ , among the 2790 species of ray-finned fish, Actinopterygii, in AmP is 0.45, so they grow by a factor of $0.45^{-3} = 11$ in weight during their adult stage, and increase considerably in reproductive output during this period. Augustine et al. (2022) found that the life-time cumulated neonate mass production $\int_{a_p}^{a_m} \dot{R}(t) W_w^b dt$ is about equal to ultimate weight W_w^∞ for many taxa; this result still applies now the AmP collection is much larger (Kooijman, 2026). While still consistent with this finding, I can here report that the weight-specific life-time cumulated neonate mass production $\omega_R = \int_{a_p}^{a_m} \dot{R}(t) W_w^b / W_w^\infty dt$ is proportional to the fraction κ_R^A of assimilation energy that is allocated to reproduction; the

proportionality constant ranges from 0.0032 for birds, to 0.631 for ray-finned fish. The proportionality is remarkable since this fraction κ_R^A is a function of the allocation fraction κ to soma and the supply stress s_s , which is numerically dominated by the maturity maintenance, and so by the maturity level at puberty. I see this as strong support for the central position of the supply–demand spectrum in animal energetics.

The median value of κ in the AmP collection of animal species is presently 0.93. It did not change much during the growth of the collection since 2009, which made (Kooijman and Lika, 2014) question a central theme of evolutionary theory: the maximization of fitness. Keeping all other parameters fixed, the value of κ that maximizes reproduction is around 0.45, details depending on the values of other parameters. If κ is small, most mobilized reserve is allocated to reproduction. However, feeding is linked to surface area, so to squared structural length, and food derived metabolites are first added to the reserve. So if growth is little during ontogeny, assimilation is little, so is the flux of mobilized reserve; the allocation to reproduction has an optimum relationship with κ . At the optimum value of κ the maximum reproduction rate can easily be 10 or 100, sometimes even more, times as high, compared with what the species actually does. Such factors make discussions about the detailed fitness measure to be optimized, such as maximum reproduction rate, life-time reproductive output or population growth rate, irrelevant. The effects of toxic compounds on the metabolic performance of individuals can frequently be well-captured by changing one or two DEB parameters in response to toxic stress, called mode of action (Gust et al., 2021; Kooijman and Bedaux, 1996; Weighman et al., 2023; Matyja, 2023; Ashauer and Jager, 2018; Forbes et al., 2017; Swain et al., 2010; Jager et al., 2007; Alda Alvarez et al., 2006), supporting the view that DEB parameters are the handles for changes in metabolism; each DEB parameter represents a single metabolic process. The tetrahedron of life sets this issue in a very different light.

The main result of this paper is the discovery of mathematical implications of a long-existing theory that is based on first principles and for which a large amount of empirical evidence has been accumulated over half a century. The tetrahedron of life relates s_s , $f_{\min}$, κ_R^A and κ ; none of these quantities relate directly to neonate size or life span. Yet it has obviously something to tell about the life-time cumulated neonate mass production and the altricial-precocial spectrum. While the minimum scaled functional response $f_{\min}$ relates to half-saturation coefficient and so to the aerobic scope, the supply stress s_s relates to the trophic position of species. The twin papers Verhille and Kooijman (2026) and Kooijman (2026) provide further discussion on this topic.

The tetrahedron of life shows that species with a high trophic position must have a high supply stress and can for this reason not allocate a large fraction of their assimilation to reproduction. This is in harmony with the fact that the predatory losses that need to be compensated by reproduction decrease towards the top of the food pyramid. The tetrahedron also shows that species at the top of the food pyramid need a feeding rate close to their maximum to reach puberty; they must have a small half-saturation coefficient and thus a high aerobic scope. These two properties might turn out to be key to the long-term dynamical stability of food pyramids, avoiding over-exploitation of resources. The tetrahedron, in combination with the observed low investment into reproduction, suggests that the avoidance of over-exploitation of resources might be of more relevance to evolution than maximizing fitness; the latter principle has been criticized more than once (Birch, 2015; Queller, 2017). The idea that over-exploitation avoidance is more important than fitness maximization for evolution is in harmony with the finding that syntrophy is its dominant nature (Kooijman and Augustine, 2026).

Reduction of over-exploitation might also (slightly) reduce the need to include social interaction to maintain biodiversity in ecosystem models (Kooijman, 2024), one of the most difficult things to achieve in ecosystem modelling. The feeding rate should not only depend on food density to avoid competitive exclusion (Kooijman, 2024); territorial

interaction is the most likely mechanism to ensure that feeding rate not only depends on food density but also on the density of conspecifics. Apart from a small half-saturation coefficient, territorial defense helps to ensure a high functional response for demand species. This suggests a link between the intensity of territorial defense and trophic position as a topic for future research. The up-regulation of assimilation by birds and mammals during the reproductive season is probably a complementary mechanism to deal with the reduced scope for feeding for demand-species; model proposals in a DEB context can be found in Kooijman (2010a).

The significance of these findings is that this coupling of traits follow from the structure of DEB theory, confirmed by data, rather than being just correlations in data. It turns out that reproduction investment and food searching behaviour, both linked to the trophic position, are in fact two sides of the same coin. This mathematical result, given some DEB assumptions, underlines the central position of the supply–demand spectrum in animal energetics.

CRedit authorship contribution statement

Sebastiaan A.L.M. Kooijman: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

There is not any conflict of interest

Acknowledgements

The author thanks all the contributors to the Add_my_Pet collection and Dina Lika for her suggestion to call κ the cornerstone of life. I also thank Mike Kearney and an anonymous reviewer for helpful comments.

Data availability

All data is available at the AmP website.

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